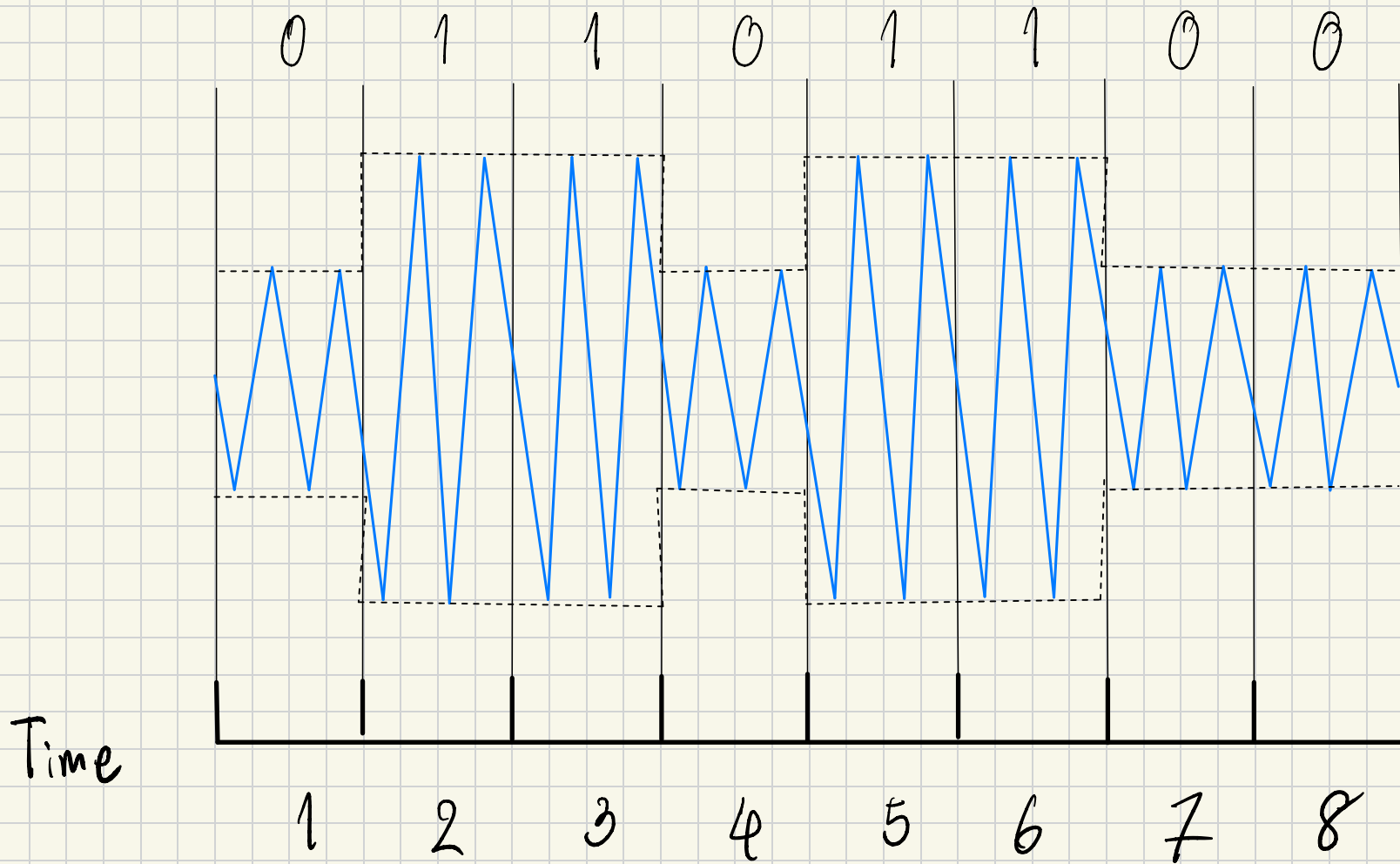


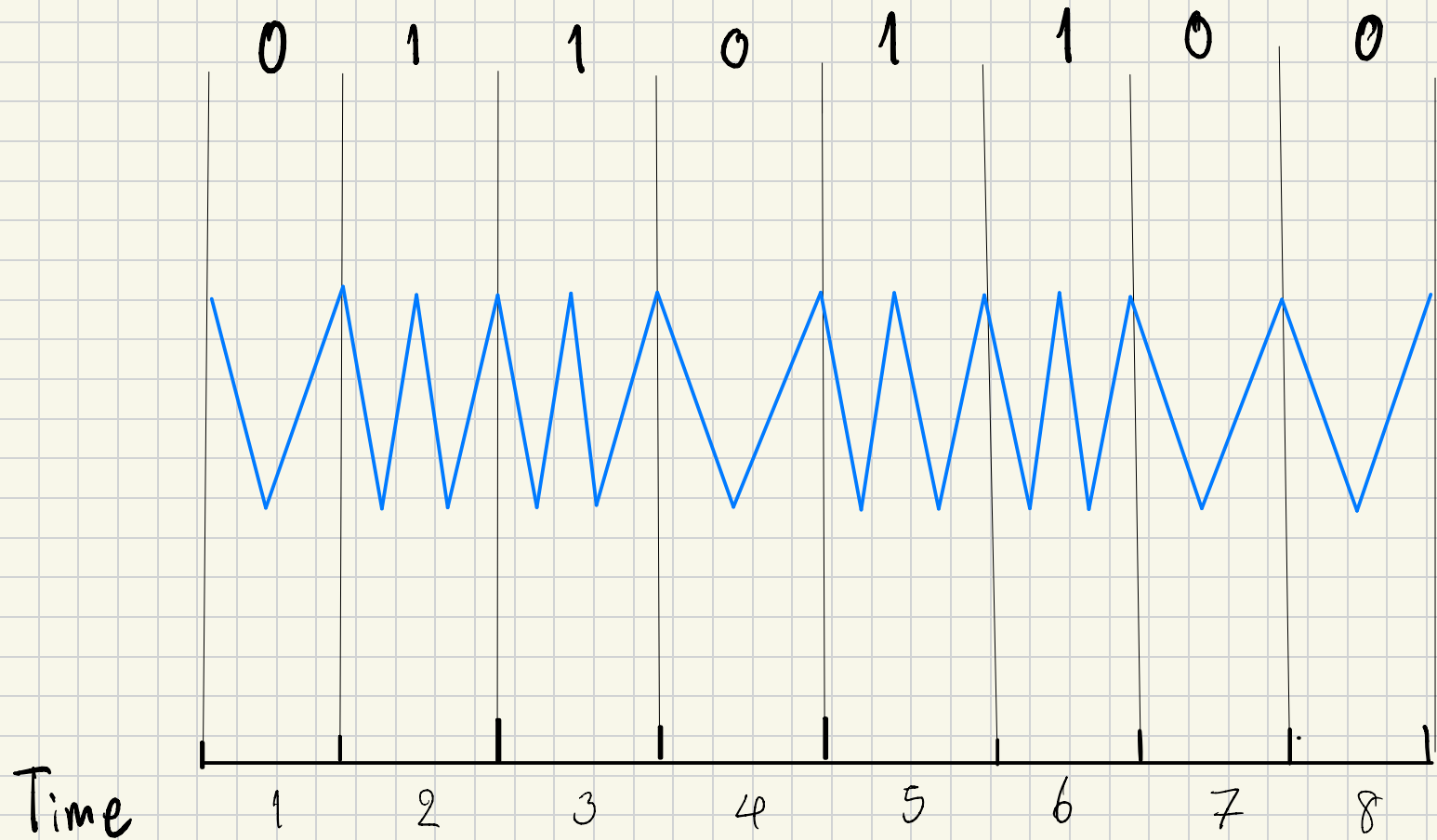
Homework 3 Exercise from chapter 3 Exercise E and F

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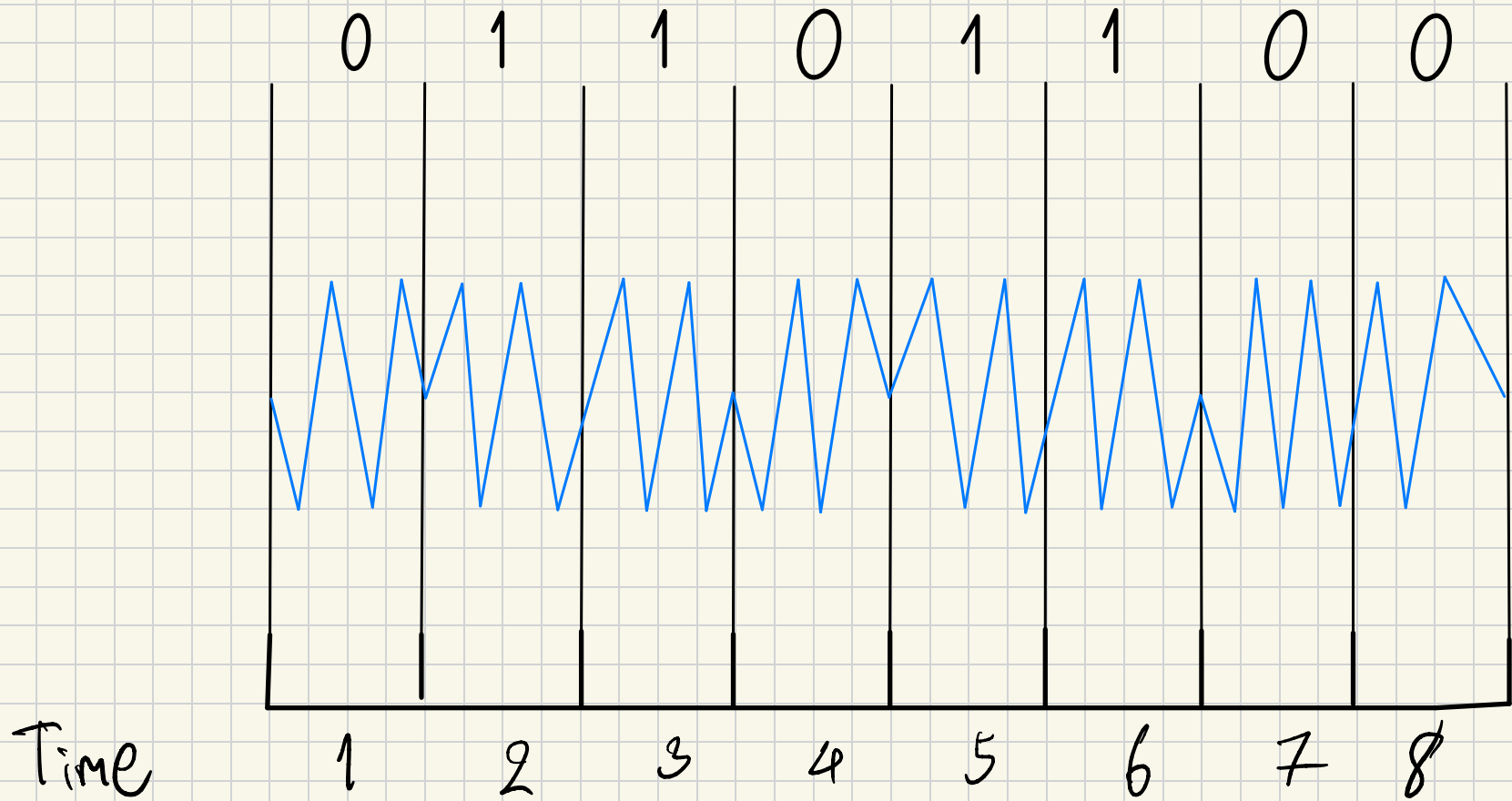
E. Draw how the bit patten 01101100 would be sent using
a. Single-bit AM



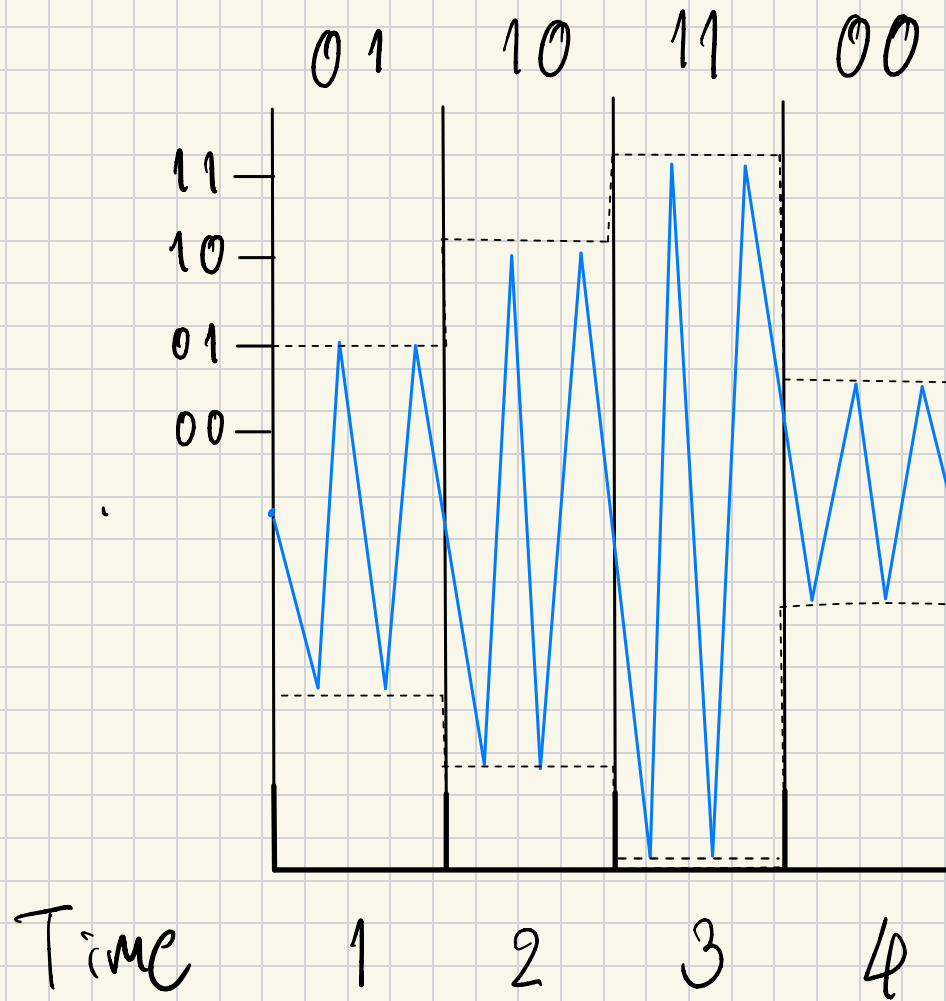
E. Draw how the bit patten 01101100 would be sent using
b. Single-bit FM



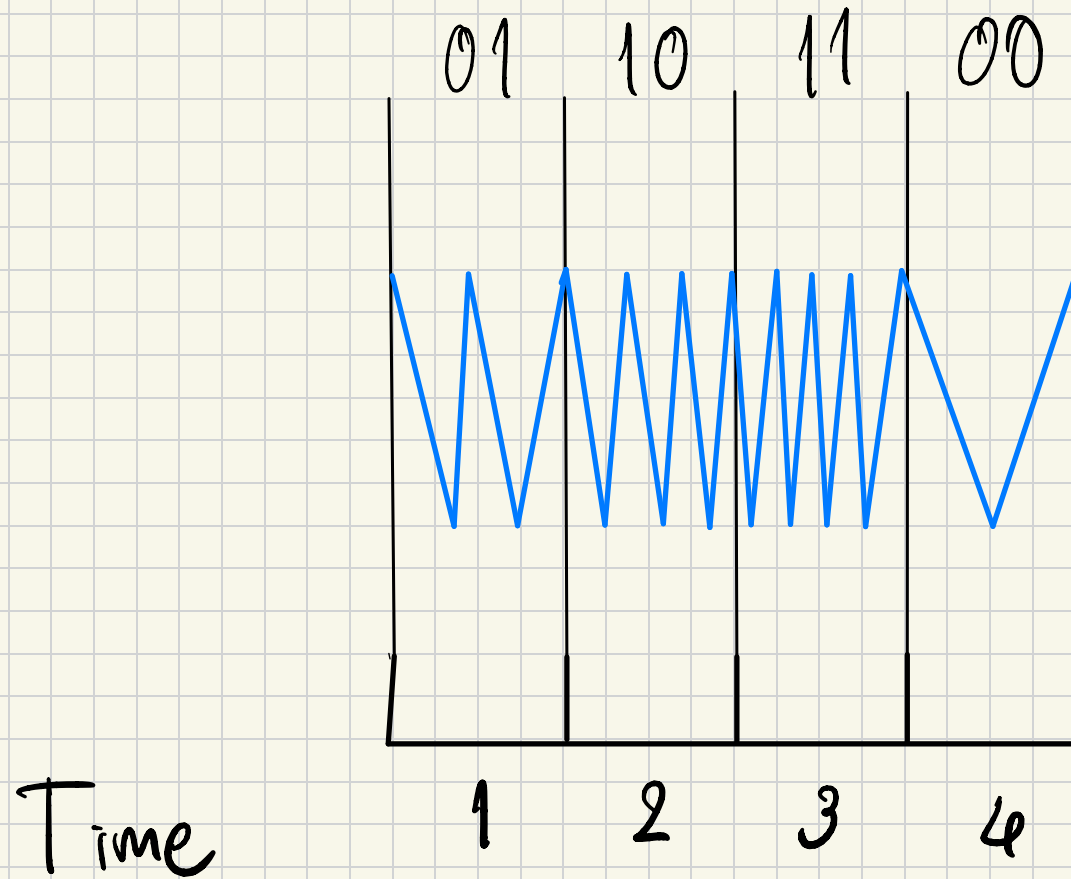
E. Draw how the bit patter 01101100 would be sent using
c. Single-bit PM



E. Draw how the bit pattern 01101100 would be sent using
d. Two-bit AM (i.e., four amplitude levels)



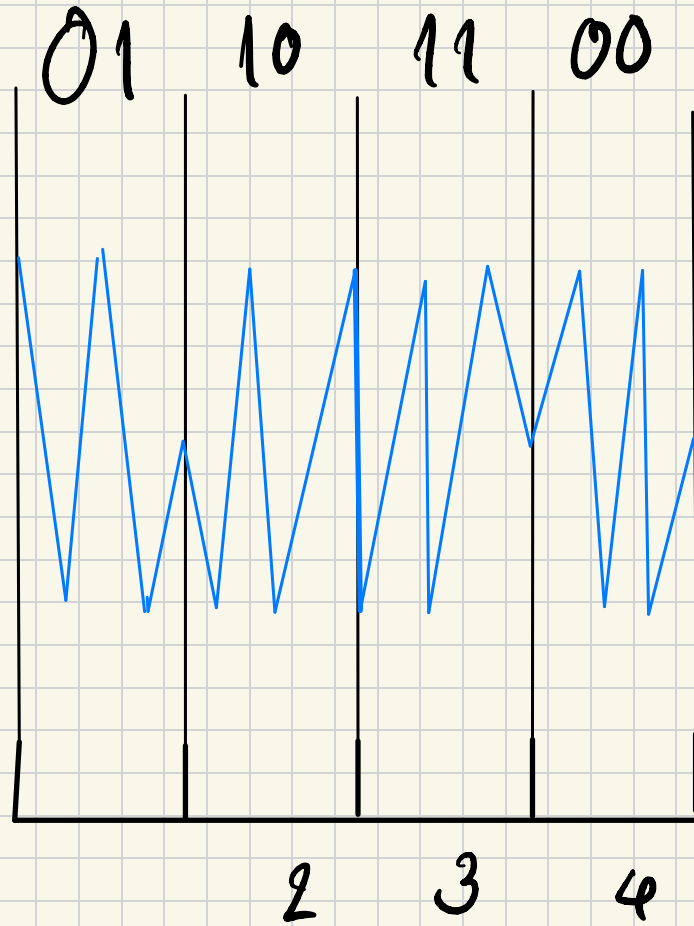
E. Draw how the bit patter 01101100 would be sent using
e.Two-bit FM (i.e., four frequencies)



E. Draw how the bit patter 01101100 would be sent using
f. Two-bit PM (i.e., different phases)

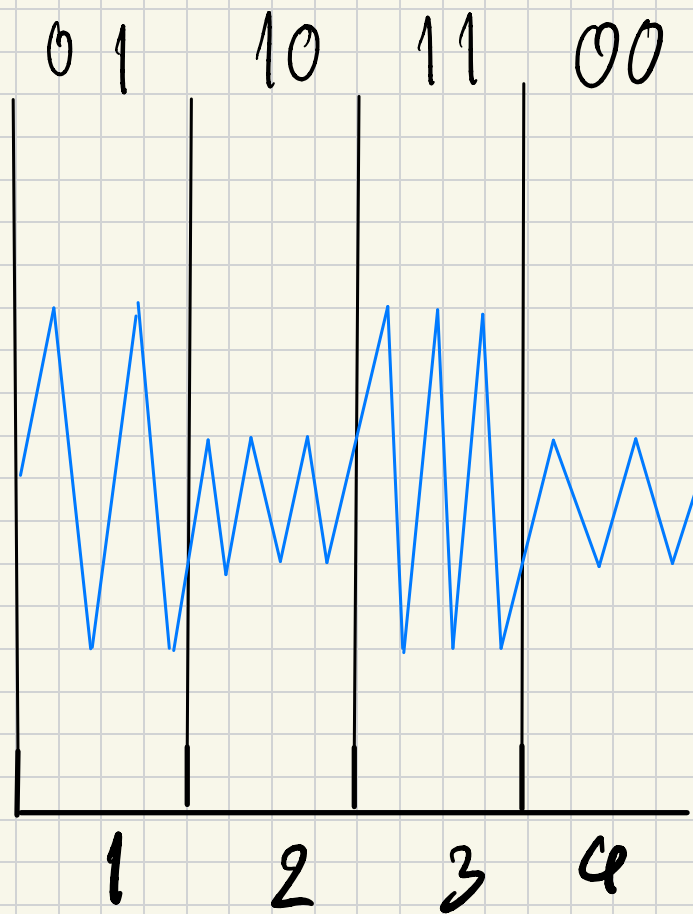
00	0°
01	90°
10	180°
11	270°

Time

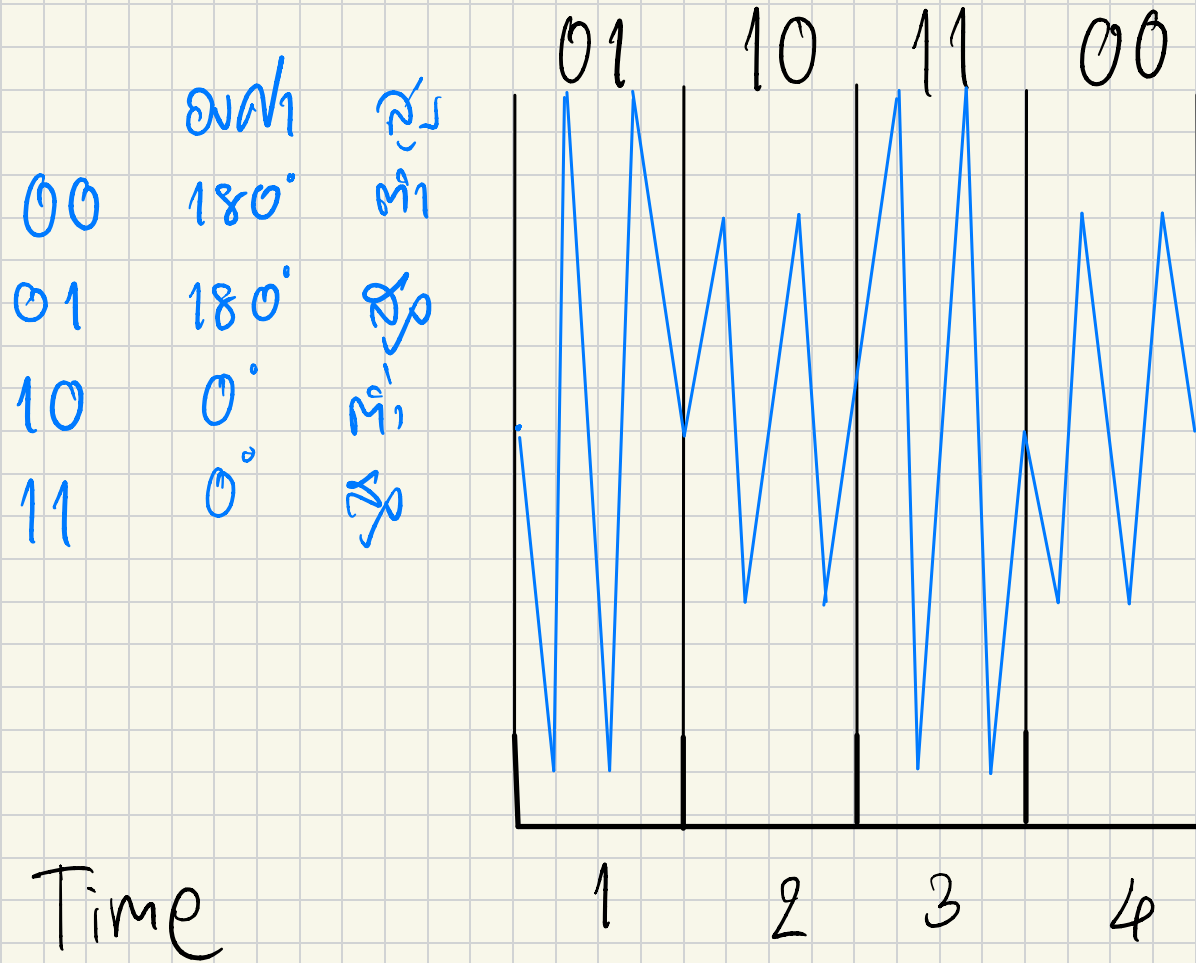


E. Draw how the bit pattern 01101100 would be sent using
g. Single-bit AM combined with single-bit FM)

	0	1
00	అధిక	తక్కు
01	అధిక	ధృ
10	తక్కు	అధిక
11	తక్కు	ధృ



E. Draw how the bit patten 01101100 would be sent using
h. Single-bit AM combined with single-bit PM)



E. Draw how the bit patter 01101100 would be sent using
i. Two-bit AM combined with Two-bit PM)

0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111

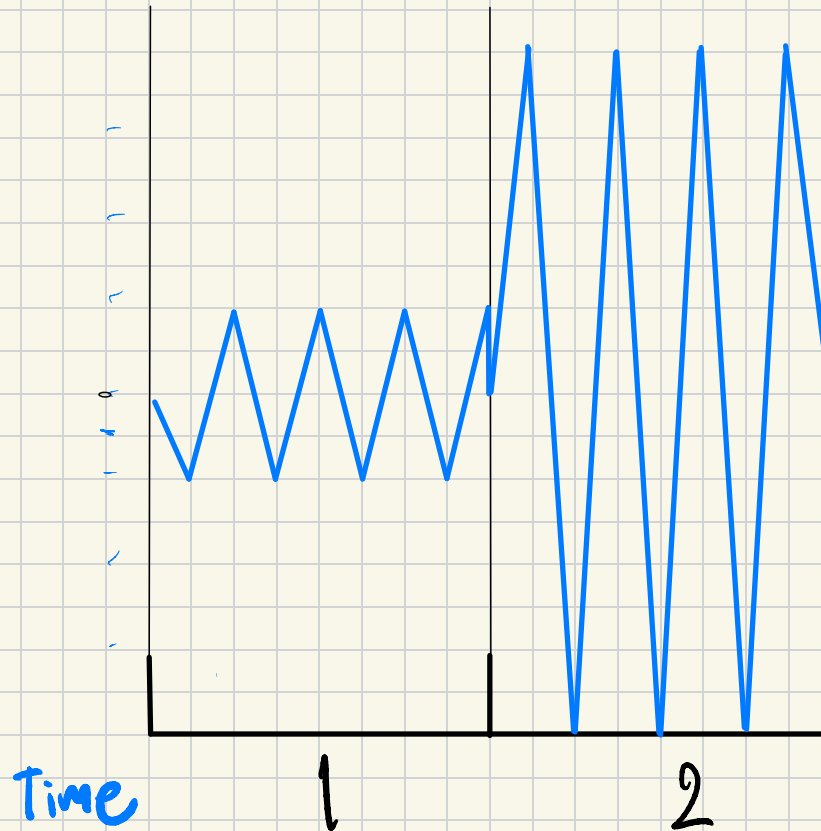
AM

PM

ต่ำมาก	0°
ต่ำมาก	90°
ต่ำมาก	180°
ต่ำมาก	270°
ต่ำ	0°
ต่ำ	90°
ต่ำ	180°
ต่ำ	270°
สูง	0°
สูง	90°
สูง	180°
สูง	270°
สูงมาก	0°
สูงมาก	90°
สูงมาก	180°
สูงมาก	270°

0110

1100



F. IF you had to download a 20-page paper of 400 k (bytes) from your professor, approximately how long would it take to transfer it over the following circuits?

Assume that control character add extra 10% to the message

Paper 400 KB

Extra 10%

Sum 440 KB

Convert 440KB to bit = $440 \times 1,024 \text{ bytes} \times 8 \text{ bits} = 3,604,480 \text{ bits}$

a. Dial-up modem at 33.6 kbps

Answer:

modem at 33.6 kbps = $33.6 \times 1000 = 33,600 \text{ bits per second}$

Dial-up modem at 33.6 kbps is $3,604,480 / 33,600 = 107.28 \text{ seconds}$

b. Cable model at 384 kbps

Answer:

Cable model at 384 kbps = $384 \times 1,000 = 384,000 \text{ bits per second}$

Dial-up modem at 33.6 kbps is $3,604,480 / 384,000 = 9.39 \text{ seconds}$

c. Cable modem at 1.5 Mbps

Answer:

Cable model at 1.5 Mbps = $1.5 \times 1,000,000 = 1,500,000 \text{ bits per second}$

Dial-up modem at 33.6 kbps is $3,604,480 / 1,500,000 = 2.40 \text{ seconds}$

d.If modem includes V.44 data compression with a 6:1 data compression ratio,what is the data rate in bits per second you would actually see in choice c ?

Answer:

Original data size = 440 KB = $440 \times 1,024 \times 8 \text{ bits} = 3,604,480 \text{ bits}$

Data compression with a 6:1 = $3,604,480 / 6 = 600,746.67 \text{ bits}$

Cable model at 1.5 Mbps = $1.5 \times 1,000,000 = 1,500,000 \text{ bits per second}$

Download time after compression. = $600,746.67 / 1,500,000 = 0.4 \text{ seconds}$

Actual data rate = $1,500,000 \times 6 = 9,000,000 \text{ bits per second or } 9 \text{ Mbps}$

Bipolar:
alternate mark
Inversion (AMI)

10011011101

